Junyan Tan (Samuel)

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Research Interests

Population Genetics; Phylodynamics; Quantitative Genetics; Large Language Model; Convolutional Neural Network.

Research Experience

- **University of Michigan, Department of Statistics** *May 2025 – Present*
  Research Assistant (Prof. Jonathan Terhorst) Ann Arbor, USA
  - Co-created `demesinfer`, a Python library for demographic inference; implemented a JAX-based optimizer that automates demographic parameter estimation (replacing manual visual calibration in earlier workflows).
  - Applied demesinfer to real datasets, including an internal unphased Michigan snake dataset, and participated in the Genomic History Inference Strategies Tournament (GHIST) using demesinfer for inferring demographic parameters.
  - Maintained and enhanced `momi3` and `phlash` population-genetic inference libraries; improved usability and stability; assisted integration of momi3 methods into demesinfer.
  - Prototyping a particle-filter approach to infer coalescent-time distributions for phylodynamics and population genetics.
- **University of Michigan School of Public Health, Department of Biostatistics** *Aug 2024 – May 2025*
  Undergraduate Research Assistant (Prof. Irina Gaynanova) Ann Arbor, USA
  - Developed a zero-shot LLM framework with Retrieval-Augmented Generation (RAG) for CGM time-series forecasting, achieving up to 30% lower error and 50% lower compute time compared to strong time-series baselines.
  - Maintained and extended the `iglu` R package for CGM analysis (bug fixes, new features, tests, documentation) and enhanced the Shiny-based visualization interface.
  - Curated and processed CGM datasets for 1,000+ individuals and coordinated analyses for a multi-institution consensus study with five external CGM software teams.
- **Academy of Mathematics and Systems Science, Chinese Academy of Sciences** *May 2024 – Aug 2024*
  Visiting Scholar / Research Intern (Prof. Qizhai Li) Beijing, China
  - Developed a novel test statistic for multimodal regression with an R implementation and theoretical guarantees.
  - Demonstrated 1.5–2× higher statistical power on simulated settings compared to prior tests; prepared an internal report.

Education

- **University of Michigan** *Aug 2025 – May 2026 (Expected)*
  M.S. Biostatistics (Accelerated) Ann Arbor, USA
- **University of Michigan** *May 2023 – May 2025*
  B.S. (Hons.) Data Science & B.S. Statistics Ann Arbor, USA
  - GPA: 3.83/4.00
  - Phi Beta Kappa (top 10% of graduating seniors)
  - Honors in Data Science
- **University of California, Santa Barbara** *Sep 2021 – Apr 2023*
  Mathematics; Statistics & Data Science (two years completed) Santa Barbara, USA
  - GPA: 4.0/4.0

Honors and Awards

- **Honors in Data Science** *May 2025*
  University of Michigan
  - Honors Thesis: *Evaluating the Zero-Shot Predictive Ability of Large Language Models for Continuous Glucose Monitoring Data.*
- **Phi Beta Kappa** *Mar 2025*
  Alpha of Michigan Chapter
  - Elected to Phi Beta Kappa (U.S. national academic honor society; top 10% of graduating seniors).
- **Best Poster Award** *Mar 2025*
  Michigan Student Symposium for Interdisciplinary Statistical Sciences
  - Awarded for the poster on LLM-based CGM time-series forecasting.

- **University Honors (2x)** *2023 – 2025*
  University of Michigan
- **Dean's Honors (4x)** *2021 – 2023*
  University of California, Santa Barbara
- **Gold Award** *2019*
  British Physics Olympiad

Publications, Preprints & Selected Works

[1] **Tan, J.** (2025). *Evaluating the Zero-Shot Predictive Ability of Large Language Models for Continuous Glucose Monitoring Data.* Undergraduate Honors Thesis, University of Michigan.

[2] Dilber, E., Liang, J., **Tan, J.**, Terhorst, J. (in preparation). *Faster inference of complex demographic models from large allele frequency spectra.* (See earlier version on bioRxiv: 10.1101/2024.03.26.586844).

[3] Kok, N., Xu, X., **Tan, J.**, et al. (work in progress). *A Standardized Consensus on Continuous Glucose Monitoring Metrics.*

[4] Xu, X., Kok, N., **Tan, J.**, et al. (2024). *Awesome-CGM: List of CGM datasets.*

[5] **Tan, J.** (2024). *A novel test statistic for multimodal regression.* Internal research report.

Talks & Presentations

- **Evaluating the Zero-Shot Predictive Ability of LLMs for CGM Time Series** *Mar 2025*
  MSSISS 2025 — Poster (Best Poster Award)
  - Presented honors-thesis results on zero-shot LLM forecasting for CGM time series; highlighted gains and the role of retrieval-augmented generation (RAG).
- **Evaluating the Zero-Shot Predictive Ability of LLMs for CGM Time Series** *Mar 2025*
  Inaugural "Look to Michigan Biostatistics" Student Research Showcase — Poster
  - Presented preliminary findings; outlined CGM datasets, evaluation metrics, and study methodology.
- **Time Series Forecasting with Large Language Models** *Dec 2024*
  University of Michigan, Department of Biostatistics — Gaynanova Group Seminar
  - Led a tutorial on LLMs for time-series forecasting in biostatistics; introduced core LLM theory and discussed applied use cases in biostatistics.

Research Projects

- **dpf — Particle Filters for Phylodynamic and Population Genetic Inference** *In development*
  Python Package, JAX, diffrax, msprime
  - Sequential Monte Carlo (particle filtering) framework to infer coalescent-time distributions from genetic / phylodynamic data under Demes-specified demographies. Prototype implemented in JAX with diffrax; includes a neural network that parameterizes the prior over demographic parameters.
- **demesinfer & momi3** *May 2025 – Present*
  Python Package, JAX, Demes, msprime
  - `demesinfer`: open-source demographic inference toolkit that supports site-frequency spectrum (SFS) and inverse instantaneous coalescence rate (IICR) based methods, implemented in Python/JAX; provides tutorials and documentation; integrates selected `momi3` methods.
  - `momi3`: library for demographic inference from the allele frequency spectrum.
  - *My contribution:* co-created `demesinfer`; implemented a JAX optimizer, wrote documentation/tutorials, and contributed patches to `momi3` while assisting its integration into `demesinfer`.
- **Gluco-LLM** *Dec 2024 – Apr 2025*
  Python, PyTorch, LLM, RAG, Hugging Face
  - A framework for forecasting continuous glucose monitoring (CGM) time series with zero-shot large language models, evaluated across multiple datasets. Utilizes retrieval-augmented generation (RAG) to improve predictive accuracy.
  - *My contribution:* sole author; repurposed a pre-trained LLM for zero-shot forecasting, built the RAG pipeline and an evaluation process; achieved 30% lower MSE and 50% lower compute vs. strong baselines.
- **Awesome-CGM** *Aug 2024 – Dec 2024*
  R, Dataset
  - Curated and processed publicly available CGM datasets; standardized metadata to create a reliable resource for benchmarking and method development.
  - *My contribution:* led data cleaning/standardization scripts and compiled documentation for contributors/users.

- **iglu** *Aug 2024 – Dec 2024*
  R/Python Package, Dashboard via R Shiny
  - Open-source R/Python package for visualizing continuous glucose monitoring (CGM) data and summarizing clinically relevant metrics.
  - *My contribution:* maintained and enhanced the package, including modifications to the Shiny dashboard, debugging new metrics, and updating documentation.
- **MMR-log** *May 2024 – Aug 2024*
  R
  - Simulation study of a novel test statistic for multimodal regression demonstrating 50–100% higher power than baseline methods.
  - *My contribution:* sole author; implemented the method and ran comparative simulations/visualizations.

Course & Club Projects

- **Analytics Dashboard: "Diabetes in the U.S."** *Jan 2024 – Apr 2024*
  Python, Visualization via Plotly Dash, Deployment on Google Cloud Platform
  - Interactive dashboard to explore U.S. diabetes indicators; integrates multiple machine learning models for subgroup analysis.
  - *My contribution:* built and deployed the app on GCP; created data processing and core visualization modules.
- **Machine Learning Project: "U.S. Demographics"** *Oct 2023 – Dec 2023*
  R, Random Forest, Lasso, Cross-Validation
  - Used multiple supervised learning methods (random forest, Lasso with cross-validation) to identify key predictors of poverty in the United States; achieved 86% test accuracy and highlighted several high-impact socioeconomic factors.
  - Co-authored the final analytical report summarizing methodology, feature importances, and policy-relevant findings.
- **Real vs. Fake** *Sep 2023 – Dec 2023*
  Python, PyTorch, CNN, TensorFlow
  - Convolutional-neural-network (CNN) model to classify photoshopped faces; trained and validated with cross-validation.
  - *My contribution:* designed the model and training loop; trained on the University of Michigan Great Lakes cluster; achieved 60–70% test accuracy on held-out data.

Selected Courses

- **Biostatistics:** Statistical Genetics, Phylodynamics, Clinical Trials.
- **Math & Statistics:** Probability & Mathematical Statistics, Real Analysis, Nonparametric Statistics, Longitudinal Analysis, Bayesian Statistics, Stochastic Processes, Generalized Linear Models, Linear Regression
- **Data Science & Computer Science:** Data Mining, Data Science & Analytics w/ Python, Data Science w/ R, Database Management Systems, Data Structures and Algorithms w/ C++

Skills

- **Programming Languages:** R, Python, C++, C, SQL, Java, JavaScript
- **Data Science & ML:** Regression/GLM, Random Forests, CNNs, LLMs, Unsupervised Learning, HMMs, MCMC
- **Tools & Platforms:** GitHub, JAX, PyTorch, TensorFlow, Hugging Face, Shiny, Plotly, GCP, MongoDB, MySQL
- **Working Languages:** English, Chinese, Japanese (fluent); French (fluent); Spanish (limited)

Volunteering & Non-Profit

- **Data Science Consultant** *Jan 2024 – Apr 2024*
  Industrial Sewing and Innovation Center (ISAIC)
  - Analytics engagement for a Detroit-based nonprofit: Python-based analyses and visualizations (seaborn, matplotlib) and a concise report used in grant applications.
  - *My contribution:* designed the analysis plan, created the visuals, and co-authored the report.
- **Education Committee** *Jan 2023 – Aug 2023*
  Michigan Data Science Team (MDST)
  - Programmed training content for student practitioners, emphasizing statistical foundations and theory in data science (regularization, cross-validation, Bayesian methods).
  - *My contribution:* led curriculum design and delivery, mentored the project team, and compiled the final report.

References

Available upon request.